LINEAR ALGEBRA & CALCULUS

(23A54101)

UNIT - V

Multiple Integrals

Topics covered:

- Double integrals and triple integrals
- Change of order of integration
- Change of variables: polar, cylindrical and spherical coordinates
- Areas by double integrals
- Volumes by double and triple integrals

Handwritten Notes - B.Tech (JNTUA R23)

1. Double Integrals

The double integral of $f(x, y)$ over a region R is the limit of $\Sigma f(x_i, y_i) \Delta A$: $\iint_R f(x, y)\, dA$. It is evaluated as an ITERATED (repeated) integral.

$$\int_{x=a}^{b} \int_{y=\varphi_1(x)}^{\varphi_2(x)} f(x, y)\, dy\, dx \quad (y\text{-limits depend on } x : \text{vertical strip})$$

$$\int_{y=c}^{d} \int_{x=\psi_1(y)}^{\psi_2(y)} f(x, y)\, dx\, dy \quad (x\text{-limits depend on } y : \text{horizontal strip})$$

- Integrate the INNER integral first, treating the outer variable as a constant.
- Constant limits (rectangle) – the order does not matter (Fubini).
- For a region R draw a rough sketch first, the limits come from the sketch.

Problem 1: evaluate $\int_0^1 \int_0^2 (x^2 + y^2)\, dy\, dx$.

$$\text{inner} : \int_0^2 (x^2 + y^2)\, dy = \left[x^2 y + y^3/3 \right]_0^2 = 2x^2 + 8/3$$

$$\boxed{\text{outer} : \int_0^1 (2x^2 + 8/3)\, dx = 2/3 + 8/3 = 10/3}$$

Problem 2: evaluate $\iint xy\, dx\, dy$ over the region bounded by the x-axis, the ordinate $x = 2a$ and the curve $x^2 = 4ay$.

For a given x, y runs from 0 to $x^2/(4a)$; x runs from 0 to $2a$.

$$\boxed{\int_0^{2a} x \left[y^2/2 \right]_0^{x^2/4a} dx = \int_0^{2a} \frac{x^5}{32a^2}\, dx = \frac{(2a)^6}{6 \cdot 32a^2} = \frac{a^4}{3}}$$

Problem 3: evaluate $\iint xy(x + y)\, dx\, dy$ over the area between $y = x^2$ and $y = x$.

The curves meet where $x^2 = x$: $x = 0, 1$. For $0 \le x \le 1$, y goes from x^2 (lower) to x (upper).

$$\int_0^1 \int_{x^2}^{x} (x^2 y + xy^2)\, dy\, dx = \int_0^1 \left[x^2 y^2/2 + xy^3/3 \right]_{x^2}^{x} dx$$

$$= \int_0^1 \left(\frac{5x^4}{6} - \frac{x^6}{2} - \frac{x^7}{3} \right) dx$$

$$= \frac{1}{6} - \frac{1}{14} - \frac{1}{24} = \frac{3}{56}$$

Problem 4: evaluate $\iint e^{x+y}\, dx\, dy$ over the triangle bounded by $x = 0$, $y = 0$, $x + y = 1$.

$$\int_0^1 \int_0^{1-x} e^{x+y}\, dy\, dx = \int_0^1 (e - e^x)\, dx = e - (e - 1) = 1$$

Problem 5: evaluate $\iint xy\, dx\, dy$ over the positive quadrant of the circle $x^2 + y^2 = a^2$.

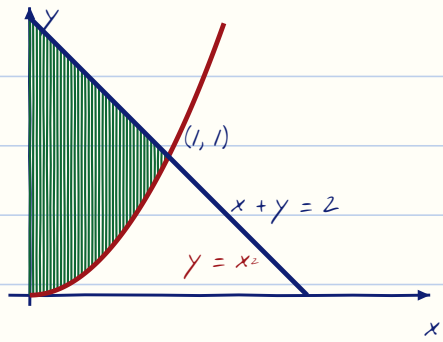
$$\int_0^a x \cdot \frac{a^2 - x^2}{2}\, dx = \frac{1}{2}\left[\frac{a^2x^2}{2} - \frac{x^4}{4} \right]_0^a = \frac{a^4}{8}$$

2. Change of Order of Integration

If the limits are such that one order is difficult (or impossible) we sketch the region and re-describe it with the other variable first.

1. Draw the region from the given limits (outer limits are numbers, inner limits are curves).
2. Draw a strip in the OTHER direction, note where it enters and leaves the region ⇒ new inner limits; the extreme values give the new outer limits.
3. If the strip enters/leaves through different curves in different parts, SPLIT the region.
4. Evaluate in the new order – the value is unchanged.

Problem 6: change the order of integration in $\int_0^1 \int_{x^2}^{2-x} xy\, dy\, dx$ and evaluate.



Region of integration:

$0 \le x \le 1$, $x^2 \le y \le 2 - x$

dy first (vertical strip)

The region lies between the parabola $y = x^2$ and the line $y = 2 - x$ for $0 \le x \le 1$ (they meet at $(1, 1)$). Draw HORIZONTAL strips:

- For $0 \le y \le 1$: x runs from 0 to $\sqrt{y}$.
- For $1 \le y \le 2$: x runs from 0 to $2 - y$.

$$I = \int_0^1 \int_0^{\sqrt{y}} xy \, dx \, dy + \int_1^2 \int_0^{2-y} xy \, dx \, dy = \int_0^1 \frac{y^2}{2} \, dy + \int_1^2 \frac{y(2-y)^2}{2} \, dy$$

$$= \frac{1}{6} + \frac{5}{24} = \frac{3}{8}$$

(Check: the original order gives $\int_0^1 x[(2-x)^2 - x^4]/2 \, dx = 3/8$ ✓.)

Problem 7: change the order and evaluate $\int_0^4 \int_{x^2/4}^{2\sqrt{x}} dy \, dx$ (the area between two parabolas).

The curves $y = x^2/4$ and $y = 2\sqrt{x}$ (i.e. $x = y^2/4 \ldots y^2 = 4x$) meet at $(0, 0)$ and $(4, 4)$. Horizontal strip: x from $y^2/4$ to $2\sqrt{y}$, $0 \le y \le 4$.

$$\int_0^4 \int_{y^2/4}^{2\sqrt{y}} dx \, dy = \int_0^4 \left(2\sqrt{y} - \frac{y^2}{4} \right) dy = \frac{32}{3} - \frac{16}{3} = \frac{16}{3}$$

Problem 8: evaluate by changing the order: $\int_0^\infty \int_x^\infty (e^{-y}/y) \, dy \, dx$.

Region: $0 \le x < \infty$, $x \le y < \infty$, i.e. $0 \le y < \infty$ and $0 \le x \le y$.

Interchanging:

$$\int_0^\infty \frac{e^{-y}}{y} \left(\int_0^y dx \right) dy = \int_0^\infty \frac{e^{-y}}{y} \cdot y \, dy = \int_0^\infty e^{-y} \, dy = 1$$

Problem 9: evaluate $\int_0^1 \int_x^1 (\sin y / y)\, dy\, dx$ (cannot be done in the given order).

($\sin y/y$ has no elementary antiderivative.) Region: $0 \le x \le 1,\ x \le y \le 1 \Rightarrow 0 \le y \le 1,\ 0 \le x \le y$.

$$\int_0^1 \frac{\sin y}{y} \left(\int_0^y dx \right) dy = \int_0^1 \sin y\, dy = 1 - \cos 1$$

Problem 10: change the order of $\int_0^a \int_{x^2/a}^{2a-x} xy\, dy\, dx$ and evaluate.

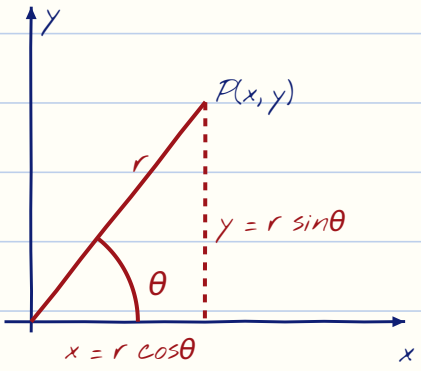
Parabola $x^2 = ay$ and line $x + y = 2a$ meet at (a, a). For $0 \le y \le a$: x from 0 to $\sqrt{(ay)}$; for $a \le y \le 2a$: x from 0 to $2a - y$.

$$I = \int_0^a \int_0^{\sqrt{(ay)}} xy\, dx\, dy + \int_a^{2a} \int_0^{2a-y} xy\, dx\, dy = \frac{a^4}{6} + \frac{5a^4}{24} = \frac{3a^4}{8}$$

3. Change of Variables – Polar Coordinates

For a double integral, a substitution $x = x(u, v),\ y = y(u, v)$ gives

$$\iint f(x, y)\, dx\, dy = \iint f(x(u, v), y(u, v))\, |J|\, du\, dv, \quad J = \frac{\partial(x, y)}{\partial(u, v)}$$



$x = r\cos\theta,\ y = r\sin\theta$

$x^2 + y^2 = r^2$

$dx\, dy = r\, dr\, d\theta$

(Jacobian = r)

$$\iint f(x, y)\, dx\, dy = \iint f(r\cos\theta, r\sin\theta) \cdot r\, dr\, d\theta$$

Use polar form when the region is a circle / sector / part of a circle or the integrand contains $x^2 + y^2$. For a circle through the origin $r = 2a\cos\theta$ use $-\pi/2 \le \theta \le \pi/2$; circle at the origin: $0 \le r \le a,\ 0 \le \theta \le 2\pi$.

Problem 11: evaluate $\iint e^{-(x^2+y^2)}\,dx\,dy$ over the first quadrant, hence find $\int_0^\infty e^{-x^2}\,dx$.

Put $x = r\cos\theta, y = r\sin\theta$:

$$I = \int_0^{\pi/2}\int_0^\infty e^{-r^2} r\,dr\,d\theta = \int_0^{\pi/2}\left[-\frac{e^{-r^2}}{2}\right]_0^\infty d\theta$$

$$= \int_0^{\pi/2}\frac{1}{2}\,d\theta = \frac{\pi}{4}$$

But $I = \left(\int_0^\infty e^{-x^2}\,dx\right)^2$ (the integral separates in x and y), so

$$\boxed{\int_0^\infty e^{-x^2}\,dx = \frac{\sqrt{\pi}}{2}}$$

Problem 12: evaluate $\int_0^a \int_0^{\sqrt{(a^2-x^2)}} (x^2+y^2)\,dy\,dx$.

Region: quarter circle $x^2+y^2 \le a^2$ in the first quadrant $\Rightarrow 0 \le r \le a, 0 \le \theta \le \pi/2$.

$$\boxed{\int_0^{\pi/2}\int_0^a r^2 \cdot r\,dr\,d\theta = \frac{a^4}{4}\cdot\frac{\pi}{2} = \frac{\pi a^4}{8}}$$

Problem 13: evaluate $\int_0^2 \int_0^{\sqrt{(2x-x^2)}} x\,dy\,dx$.

$y^2 \le 2x - x^2 \Rightarrow (x-1)^2 + y^2 \le 1$: upper half of the circle with centre $(1, 0)$, $r = 2\cos\theta$, $0 \le \theta \le \pi/2$.

$$\int_0^{\pi/2}\int_0^{2\cos\theta} (r\cos\theta)\, r\,dr\,d\theta = \int_0^{\pi/2}\cos\theta\cdot\frac{8\cos^3\theta}{3}\,d\theta$$

$$\boxed{= \frac{8}{3}\int_0^{\pi/2}\cos^4\theta\,d\theta = \frac{8}{3}\cdot\frac{3\pi}{16} = \frac{\pi}{2}}$$

Problem 14: evaluate $\iint dx\,dy/(1+x^2+y^2)$ over the circle $x^2+y^2 \le a^2$.

$$\int_0^{2\pi}\int_0^{a}\frac{r}{1+r^2}\,dr\,d\theta = 2\pi\cdot\frac{1}{2}\log(1+a^2) = \pi\log(1+a^2)$$

Problem 15 (general substitution): evaluate $\iint (x^2+y^2)\,dx\,dy$ over the region $|x|+|y|\le 1$ using $u=x+y$, $v=x-y$.

$x=(u+v)/2$, $y=(u-v)/2$; $J=\partial(x,y)/\partial(u,v)=-1/2 \Rightarrow |J|=1/2$. Region: $-1\le u\le 1$, $-1\le v\le 1$. $x^2+y^2=(u^2+v^2)/2$.

$$I=\int_{-1}^{1}\int_{-1}^{1}\frac{u^2+v^2}{2}\cdot\frac{1}{2}\,du\,dv=\frac{1}{4}\left[2\cdot\frac{2}{3}\cdot 2\right]=\frac{2}{3}$$

4. Triple Integrals

$\iiint_V f(x,y,z)\,dV=\int_{x1}^{x2}\int_{y_1(x)}^{y_2(x)}\int_{z_1(x,y)}^{z_2(x,y)} f\,dz\,dy\,dx$. Integrate the INNERMOST variable first. If $f=1$ the integral gives the VOLUME of V.

Problem 16: evaluate $\int_0^a\int_0^b\int_0^c (x^2+y^2+z^2)\,dz\,dy\,dx$.

$$=\int_0^a\int_0^b\left[(x^2+y^2)c+\frac{c^3}{3}\right]dy\,dx$$

$$=\int_0^a\left[x^2bc+\frac{b^3c}{3}+\frac{bc^3}{3}\right]dx=\frac{abc(a^2+b^2+c^2)}{3}$$

Problem 17: evaluate $\iiint x\,y\,z\,dx\,dy\,dz$ over the tetrahedron $x\ge 0$, $y\ge 0$, $z\ge 0$, $x+y+z\le 1$.

$$\int_0^1\int_0^{1-x}\int_0^{1-x-y} xyz\,dz\,dy\,dx=\int_0^1\int_0^{1-x} xy\,\frac{(1-x-y)^2}{2}\,dy\,dx$$

$$=\int_0^1 x\,\frac{(1-x)^4}{24}\,dx=\frac{1}{720}$$

Problem 18: find $\iiint x\,dV$ and the volume of the same tetrahedron $x + y + z \le 1$ (first octant).

Volume $= \int_0^1 \int_0^{1-x} (1 - x - y)\,dy\,dx = \int_0^1 (1 - x)^2/2\,dx = 1/6$. $\iiint x\,dV = \int_0^1 x(1 - x)^2/2\,dx = 1/24$.

Problem 19: evaluate $\iiint (x + y + z)\,dV$ over the cube $0 \le x, y, z \le 1$.

By symmetry each of x, y, z contributes $\iiint x\,dV = 1/2$: total $= 3/2$.

Cylindrical coordinates: $x = r\cos\theta, y = r\sin\theta, z = z$; $dV = r\,dr\,d\theta\,dz$. Use for cylinders, cones, paraboloids with z-axis symmetry.

Spherical coordinates: $x = r\sin\theta\cos\varphi, y = r\sin\theta\sin\varphi, z = r\cos\theta$; $dV = r^2\sin\theta\,dr\,d\theta\,d\varphi$ ($0 \le r$, $0 \le \theta \le \pi$, $0 \le \varphi \le 2\pi$). Use for spheres, cones about the z-axis.

System	Variables	dV	$x^2 + y^2 + z^2$
Cartesian	x, y, z	$dx\,dy\,dz$	$x^2 + y^2 + z^2$
Cylindrical	r, θ, z	$r\,dr\,d\theta\,dz$	$r^2 + z^2$
Spherical	r, θ, φ	$r^2\sin\theta\,dr\,d\theta\,d\varphi$	r^2

Problem 20: evaluate $\iiint z\,dV$ over the cylinder $x^2 + y^2 \le a^2$, $0 \le z \le h$ (cylindrical coordinates).

$$\int_0^{2\pi}\int_0^{a}\int_0^{h} z \cdot r\,dz\,dr\,d\theta = 2\pi \cdot \frac{a^2}{2} \cdot \frac{h^2}{2} = \frac{\pi a^2 h^2}{2}$$

Problem 21: evaluate $\iiint (x^2 + y^2 + z^2)\,dV$ over the sphere $x^2 + y^2 + z^2 \le 1$.

$$\int_0^{2\pi}\int_0^{\pi}\int_0^{1} r^2 \cdot r^2\sin\theta\,dr\,d\theta\,d\varphi = \frac{1}{5} \cdot 2 \cdot 2\pi = \frac{4\pi}{5}$$

Problem 22: evaluate $\iiint xyz\,dV$ over the positive octant of the sphere $x^2 + y^2 + z^2 \le a^2$.

$$\int_0^a r^5\,dr \cdot \int_0^{\pi/2} \sin^3\theta\cos\theta\,d\theta \cdot \int_0^{\pi/2} \sin\varphi\cos\varphi\,d\varphi$$

$$= \frac{a^6}{6} \cdot \frac{1}{4} \cdot \frac{1}{2} = \frac{a^6}{48}$$

5. Areas by Double Integrals

$$\text{Area of } R = \iint_R dx\,dy \quad (\text{polar}: \iint r\,dr\,d\theta)$$

Problem 23: find the area bounded by the parabolas $y^2 = 4ax$ and $x^2 = 4ay$.

They meet at $(0, 0)$ and $(4a, 4a)$. For $0 \le x \le 4a$: y from $x^2/4a$ to $2\sqrt{(ax)}$.

$$A = \int_0^{4a} \left(2\sqrt{a}\,\sqrt{x} - \frac{x^2}{4a} \right) dx$$

$$= \left[\frac{4\sqrt{a}}{3} x^{3/2} - \frac{x^3}{12a} \right]_0^{4a} = \frac{32a^2}{3} - \frac{16a^2}{3} = \frac{16a^2}{3}$$

Problem 24: find the area of the ellipse $x^2/a^2 + y^2/b^2 = 1$.

$A = 4\int_0^a \int_0^{(b/a)\sqrt{(a^2-x^2)}} dy\,dx = (4b/a)\int_0^a \sqrt{(a^2 - x^2)}\,dx = (4b/a)(\pi a^2/4)$:

$$A = \pi a b$$

Problem 25: find the area of the cardioid $r = a(1 + \cos\theta)$.

$$A = 2\int_0^{\pi} \int_0^{a(1+\cos\theta)} r\,dr\,d\theta = a^2 \int_0^{\pi} (1 + \cos\theta)^2\,d\theta$$

$$= a^2 \left(\pi + 0 + \pi/2 \right) = \frac{3\pi a^2}{2}$$

Problem 26: find the area enclosed by $y = x$ and $y = x^2$. And the area of the circle $r = a$.

$\int_0^1 (x - x^2)\,dx = 1/2 - 1/3 = 1/6$. Circle: $\int_0^{2\pi} \int_0^a r\,dr\,d\theta = 2\pi \cdot a^2/2 = \pi a^2$.

Problem 27: find the total area of the lemniscate $r^2 = a^2 \cos 2\theta$.

$$A = 4\int_0^{\pi/4}\int_0^{a\sqrt{\cos 2\theta}} r\,dr\,d\theta = 2a^2\int_0^{\pi/4}\cos 2\theta\,d\theta = 2a^2\cdot\frac{1}{2} = a^2$$

Problem 28: find the area inside the circle $r = 2a\cos\theta$ and outside $r = a$.

They meet where $2a\cos\theta = a \Rightarrow \theta = \pm\pi/3$.

$$A = 2\int_0^{\pi/3}\int_a^{2a\cos\theta} r\,dr\,d\theta = \int_0^{\pi/3}(4a^2\cos^2\theta - a^2)\,d\theta$$

$$= a^2[\theta + \sin 2\theta]_0^{\pi/3}$$

$$= a^2\left(\frac{\pi}{3} + \frac{\sqrt{3}}{2}\right)$$

6. Volumes by Double and Triple Integrals

$$\text{Volume} = \iint_R (z_{upper} - z_{lower})\,dx\,dy$$

$$\text{Volume} = \iiint_V dx\,dy\,dz$$

Problem 29: find the volume of the sphere of radius a (double integral).

$V = 2\iint\sqrt{(a^2 - x^2 - y^2)}\,dx\,dy$ over the circle $x^2 + y^2 \le a^2$ (upper half-sphere doubled). Polar:

$$V = 2\int_0^{2\pi}\int_0^a \sqrt{(a^2 - r^2)}\,r\,dr\,d\theta = 2\cdot 2\pi\cdot\frac{a^3}{3} = \frac{4\pi a^3}{3}$$

Problem 30: find the volume of the solid bounded by the paraboloid $z = x^2 + y^2$ and the plane $z = 4$.

$$V = \iint (4 - r^2)\, r\, dr\, d\theta = 2\pi \left[2r^2 - \frac{r^4}{4} \right]_0^2 = 2\pi (8 - 4) = 8\pi$$

Problem 31: find the volume of the tetrahedron bounded by the coordinate planes and $x + y + z = 1$ using a double integral.

$V = \iint (1 - x - y)\, dy\, dx$ over $0 \le x \le 1, 0 \le y \le 1 - x = \int_0^1 (1 - x)^2/2\, dx = 1/6$. (For $x/a + y/b + z/c = 1$: $V = abc/6$.)

Problem 32: find the volume of the solid common to the cylinders $x^2 + y^2 = a^2$ and $x^2 + z^2 = a^2$.

In the first octant $z = \sqrt{(a^2 - x^2)}$ over the quarter-circle $0 \le y \le \sqrt{(a^2 - x^2)}$, $0 \le x \le a$; total = 8 × first octant.

$$V = 8 \int_0^a \int_0^{\sqrt{(a^2 - x^2)}} \sqrt{(a^2 - x^2)}\, dy\, dx$$

$$= 8 \int_0^a (a^2 - x^2)\, dx = 8 \cdot \frac{2a^3}{3} = \frac{16a^3}{3}$$

Problem 33: find the volume of the cone $z = \sqrt{(x^2 + y^2)}$ cut off by the plane $z = h$ (cylindrical coordinates).

$$V = \int_0^{2\pi} \int_0^h \int_r^h r\, dz\, dr\, d\theta = 2\pi \int_0^h r(h - r)\, dr = 2\pi \frac{h^3}{6} = \frac{\pi h^3}{3}$$

Problem 34: find the volume of the ellipsoid $x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$ (triple integral).

Put $x = a\,u, y = b\,v, z = c\,w$ (Jacobian = abc): the ellipsoid becomes the unit sphere $u^2 + v^2 + w^2 \le 1$ whose volume is $4\pi/3$.

$$V = abc \iiint_{\text{unit sphere}} du\, dv\, dw = \frac{4\pi abc}{3}$$

Problem 35: find the volume of the solid bounded by $z = 0$ and $z = 1 - x^2 - y^2$.

$$V = \int_0^{2\pi} \int_0^1 (1 - r^2)\, r\, dr\, d\theta = 2\pi \left[\frac{1}{2} - \frac{1}{4} \right] = \frac{\pi}{2}$$

Problem 36: find the volume of the part of the sphere $x^2 + y^2 + z^2 = a^2$ lying inside the cylinder $x^2 + y^2 = ax$.

The cylinder is the circle $r = a\cos\theta$ in polar form; the sphere gives $z = \pm\sqrt{(a^2 - r^2)}$. By symmetry $V = 4 \times$ (first quadrant part):

$$V = 4 \int_0^{\pi/2} \int_0^{a\cos\theta} \sqrt{(a^2 - r^2)}\, r\, dr\, d\theta$$

$$= \frac{4a^3}{3} \int_0^{\pi/2} (1 - \sin^3\theta)\, d\theta = \frac{4a^3}{3} \left(\frac{\pi}{2} - \frac{2}{3} \right)$$

$$V = \frac{2a^3}{9} (3\pi - 4)$$

Summary of useful results:

Quantity	Value
Volume of sphere / hemisphere	$4\pi a^3/3$ / $2\pi a^3/3$
Volume of cylinder (radius a, height h)	$\pi a^2 h$
Volume of cone (radius a, height h)	$\pi a^2 h/3$
Area of circle / ellipse	πa^2 / πab
Jacobians	polar r; cylindrical r; spherical $r^2 \sin\theta$
$\int_0^\infty e^{-x^2}\, dx$	$\sqrt{\pi}/2$

Important Questions - Unit V

1. Evaluate double integrals over given regions (rectangle, triangle, curves).
2. Change the order of integration and evaluate (problems 6-10).
3. Evaluate integrals by changing to polar coordinates.
4. Evaluate triple integrals, use cylindrical and spherical coordinates.
5. Find areas by double integration (parabolas, ellipse, cardioid, lemniscate).
6. Find volumes by double / triple integrals (sphere, paraboloid, cone, cylinders, ellipsoid, tetrahedron).